Andrew S. Battenberg

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Education

University of Wisconsin–Madison

Bachelor of Science, Mechanical Engineering,
Certificate in Environmental Studies
Expected Graduation: May 2026

Experience

Department of Transportation (DOT), Madison WI
Engineering Intern, June 2024 - August 2024

- Reviewed and analyzed engineering contracts for compliance with state standards and document retention requirements
- Participated in field site assessments to verify roadway alignment, shoulder width, and spacing against design specifications
- Tracked bridge projects using a state evaluation program to assess constructability and cost-effectiveness of proposed improvements

Great Lakes Bioenergy Research Center, Wisconsin Energy Institute, Madison WI
Lab Assistant, Noguera Lab, January 2024-May 2024

- Supported research focused on improving efficiency and cost-effectiveness of bioplastic precursor production
- Conducted and monitored multiple bioreactor experiments, independently managing several test runs
- Gained hands-on experience with laboratory equipment operation, chemical procurement procedures, and lab safety protocols

Projects

Mohr's Circle Learning Tool — University of Wisconsin–Madison Senior Design Project

- Prepared surfaces, applied, and soldered strain gauges into a Wheatstone bridge circuit as part of a physical deformation measurement tool
- Developed wiring schematics for strain gauge integration, documenting the electronic design for the hardware assembly
- Conducted electronics debugging using diagnostic software tools to identify and isolate defective components in the circuit

Custom BLE Macro Keyboard for CAD and 3D Modeling — Independent Project

- Designed and built a Bluetooth Low Energy (BLE) macro keyboard using an Arduino-compatible microcontroller to optimize 3D modeling workflows
- Modeled and fabricated the mechanical enclosure and layout using SolidWorks / Fusion 360, iterating based on ergonomics and usability testing

Remote Beach Closure Sign — University of Wisconsin–Madison Team Project

- Assisted in the circuit design and manufacturing of a remotely controlled beach closure sign using an Arduino with a cellular (SIM) shield
- Implemented and tested a system that received SMS messages to update sign content remotely, enabling real-time public notification

Skills

SolidWorks, Fusion 360, MATLAB, Python, AutoCAD, Excel, Arduino, C++, EES